



Internship Student Portal

Complete Development Plan

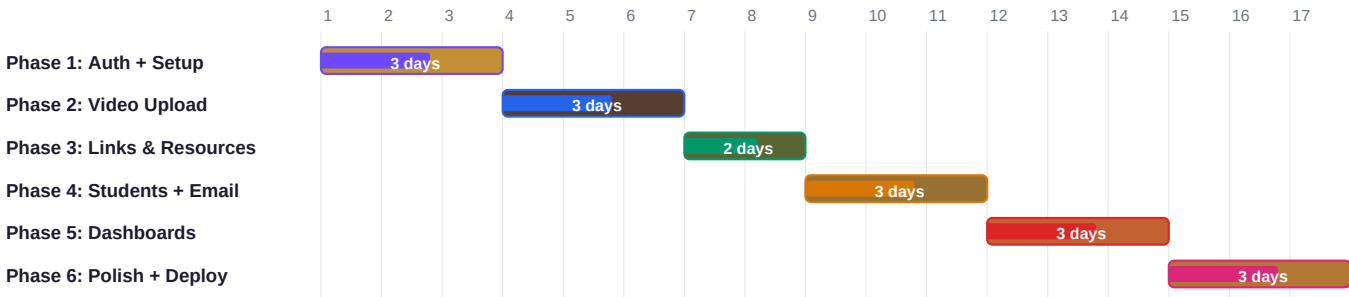
Video Upload · Resource Links · Student Email Accounts · Admin Dashboard

17	6	47+	20+
Days Total	Phases	Tasks	API Routes

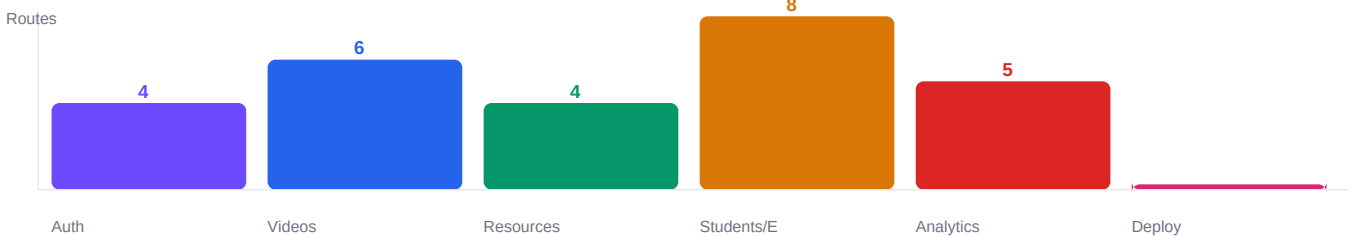
Phase	Title	Days	Tasks	Output
1	Project Setup & Authentication	1–3	8	Working login system — admin aur student dono ke liye alag a
2	Video Upload & Management System	4–6	8	Admin videos upload kar sakta hai. Students dekh sakte hain.
3	Links & Resources Section	7–8	6	Admin ek hi jagah se sab links manage karta hai. Students fi
4	Student Management & Email System	9–11	8	Admin portal se hi student accounts banata aur emails bhejta
5	Dashboards & Analytics	12–14	7	Admin ko complete visibility. Students ko progress + motivat
6	Polish, Security & Deployment	15–17	8	Live working portal — duniya mein koi bhi access kar sakta h

Project Overview — Graphs & Charts

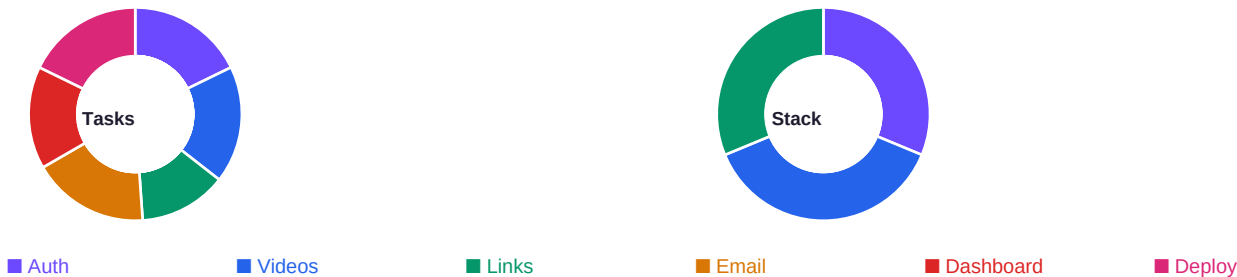
17-Day Development Timeline (Gantt Chart)



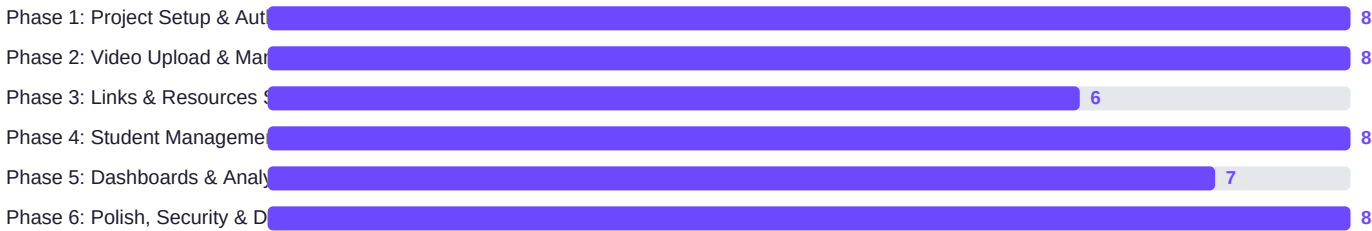
API Routes Per Phase



Task Distribution & Tech Stack Split



Task Count Per Phase (Horizontal Bar)



Phase 1 ■ Project Setup & Authentication

Days 1–3

Poori foundation yahan banegi — project structure, database connection, aur complete JWT auth system admin + student roles ke saath.

WHY PEHLE?

Bina auth ke koi cheez secure nahi hogi. Isliye ye pehle banate hain.

TASKS — 8 kaam is phase mein

1. Vite React project create karo — LearnKins purple+orange theme setup
2. Express backend: routes/ controllers/ models/ middleware/ folder structure
3. MongoDB Atlas cluster connect karo + .env file mein URI daalo
4. User schema: name, email, password(hashed), role(admin/student), batch, active
5. Admin-only route: naye students ke liye specific email se account banao
6. Student login: email + password → JWT token milega
7. Auth middleware: har route pe role check (admin ya student)
8. Forgot password: Nodemailer se reset email bhejo

API ROUTES — Is phase mein ye routes banaoge

POST	/api/auth/login
POST	/api/auth/register
GET	/api/auth/me
POST	/api/auth/reset-password

OUTPUT

Working login system — admin aur student dono ke liye alag alag access

Phase 2 ■ Video Upload & Management System

Days 4–6

Admin Cloudinary pe videos upload karta hai. Students dekhte hain. Har student ka watch progress track hota hai.

WHY PEHLE?

Ye portal ki main feature hai. Videos properly upload + store + stream hone chahiye.

TASKS — 8 kaam is phase mein

1. Cloudinary account setup + CLOUDINARY_NAME/KEY/SECRET .env mein daalo
2. Multer middleware: multipart/form-data file upload handle karo
3. Video schema: title, desc, cloudinaryUrl, thumbnail, day, module, batch[], order
4. Admin upload page: drag & drop UI + real-time upload progress bar
5. Admin manage: videos edit karo (title/desc), reorder karo, delete karo
6. Student video library: day aur module ke hisaab se grouped view
7. React Player: custom controls — play/pause/seek/speed/fullscreen
8. Progress tracking: watchedPercent DB mein save karo per student per video

API ROUTES — Is phase mein ye routes banaoge

POST	/api/videos/upload
GET	/api/videos
GET	/api/videos/:id
PUT	/api/videos/:id
DELETE	/api/videos/:id
POST	/api/progress/update

OUTPUT

Admin videos upload kar sakta hai. Students dekh sakte hain. Progress save hota hai.

Phase 3 ■ Links & Resources Section

Days 7–8

Admin curated links share karta hai — documentation, tools, articles — batch-wise targeting ke saath.

WHY PEHLE?

Students ko sab kuch ek jagah milna chahiye. Alag links dhundhne mein time waste hota hai.

TASKS — 6 kaam is phase mein

1. Resource schema: title, url, type(doc/tool/article/repo), category, pinned, batch[]
2. Admin panel: links add/edit/delete + category assign karo
3. Batch targeting: specific links sirf specific batches ko dikhao
4. Student resources page: tabs se filter karo (All/Docs/Tools/Articles/Repos)
5. Link preview cards: URL se favicon + title + description auto-fetch
6. Pin important links top pe — urgent resources ke liye

API ROUTES — Is phase mein ye routes banaoge

POST	/api/resources
GET	/api/resources
PUT	/api/resources/:id
DELETE	/api/resources/:id

OUTPUT

Admin ek hi jagah se sab links manage karta hai. Students filtered view mein dekhte hain.

Phase 4 ■ Student Management & Email System

Days 9–11

Admin student accounts banata hai, batches manage karta hai, aur portal se seedha emails bhejta hai.

WHY PEHLE?

Ye admin ki main power feature hai — student onboarding se leke communication tak sab yahan.

TASKS — 8 kaam is phase mein

1. Student creation form: name, email, batch, joining date → auto temp password generate
2. Welcome email auto-send on account creation (HTML email template)
3. Batch management: create, rename, delete batches + students assign karo
4. Student list: name/email se search, batch/status se filter, paginated
5. Account activate/deactivate: bina delete kiye login block karo
6. Bulk email composer: rich text editor, send to all ya specific batch
7. 3 email templates: Welcome, Weekly Reminder, Custom Announcement
8. Email history log: kya bheja, kisko, kab + delivery status

API ROUTES — Is phase mein ye routes banaoge

POST	/api/students/create
GET	/api/students
PUT	/api/students/:id
DELETE	/api/students/:id
POST	/api/email/send
GET	/api/email/logs
POST	/api/batches
GET	/api/batches

OUTPUT

Admin portal se hi student accounts banata aur emails bhejta hai — koi alag tool nahi.

Phase 5 ■ Dashboards & Analytics

Days 12–14

Admin ke liye sab stats. Students ke liye personal progress. LearnKins ke gamified badges bhi.

WHY PEHLE?

Data dikhna zaroori hai — admin ko pata chale kya ho raha hai, students ko motivation mile.

TASKS — 7 kaam is phase mein

1. Admin dashboard: students/videos/batches/emails ka total count + recent logins
2. Per-batch progress: har batch mein har student ka completion %
3. Per-student drill-down: kaun sa video dekha, kitna %, kitna time
4. Student dashboard: aaj ka content, overall progress bar, recent resources
5. Recharts: bar chart (videos/day), line chart (weekly activity), doughnut (completion)
6. Gamified badges: First Video, 50% Complete, All Done — student profile pe
7. Announcement/notice board: admin pinned notices seedha dashboard pe

API ROUTES — Is phase mein ye routes banaoge

GET	/api/analytics/admin
GET	/api/analytics/batch/:id
GET	/api/analytics/student/:id
POST	/api/announcements
GET	/api/announcements

OUTPUT

Admin ko complete visibility. Students ko progress + motivation milti hai.

Phase 6 ■ Polish, Security & Deployment

Days 15–17

Mobile responsive, error handling, security hardening, aur free tier pe live deployment.

WHY PEHLE?

Agar app slow, broken ya insecure hai toh sab mehenat waste. Deploy karna zaroori hai.

TASKS — 8 kaam is phase mein

1. Full mobile responsive: admin panel + student portal dono mobile pe kaam karein
2. Loading skeleton screens: har data-fetching page pe loading state
3. Global error boundary + toast notifications (success/error/info)
4. Frontend form validation + backend input sanitization
5. Rate limiting on /api/auth/* routes (express-rate-limit package)
6. Backend deploy to Render.com — sab env variables dashboard mein set karo
7. Frontend deploy to Netlify — build command + _redirects file for SPA
8. Admin user guide + README.md GitHub ke liye

OUTPUT

Live working portal — duniya mein koi bhi access kar sakta hai URL se.

Pages & Screens — Portal Ka Complete Map

ADMIN PANEL — 9 Screens (Role: admin only)			
Icon	Screen	Route	Description
	Login	/login	Admin secure login page
	Dashboard	/admin	Stats, announcements, overview
	Upload Video	/admin/upload	Drag & drop video uploader
	Manage Videos	/admin/videos	Edit, reorder, delete videos
	Add Links	/admin/resources	Curated resource link manager
	Create Student	/admin/students/new	New account + auto email
	Student List	/admin/students	Search, filter, manage accounts
	Send Email	/admin/email	Bulk email composer
	Batch Manager	/admin/batches	Create & manage student batches

STUDENT PORTAL — 6 Screens (Role: student only)			
Icon	Screen	Route	Description
	Login	/login	Student login page
	My Dashboard	/student	Today's content + progress
	Video Library	/student/videos	All videos by day/module
	Watch Video	/student/videos/:id	Player with progress tracking
	Resources	/student/resources	Filtered links & docs
	My Progress	/student/progress	Badges + completion charts

Database Models — MongoDB Mongoose Schemas

User Model

_id	ObjectId
name	String, required
email	String, unique
password	String (bcrypt)
role	enum: admin student
batch	ObjectId → Batch
active	Boolean, default true
createdAt	Date, auto

Video Model

_id	ObjectId
title	String, required
cloudinaryUrl	String
thumbnail	String
day	Number
module	String
batch	[ObjectId]
order	Number
duration	Number (sec)

Resource Model

_id	ObjectId
title	String, required
url	String
type	enum: doc tool article repo
category	String
pin	Boolean
batch	[ObjectId]

Batch Model

_id	ObjectId
name	String, required
startDate	Date
endDate	Date
active	Boolean, default true

Progress Model

_id	ObjectId
student	ObjectId → User
video	ObjectId → Video
watchedPercent	Number 0-100
completed	Boolean
lastWatched	Date

EmailLog Model

_id	ObjectId
sentBy	ObjectId → User
recipients	[String]
subject	String
template	String
sentAt	Date
status	enum: sent failed

MODEL RELATIONSHIPS

From	Relation	To	Meaning
User	→ batch →	Batch	1 student belongs to 1 batch

Video	→ batch[] →	Batch	1 video visible to multiple batches
Resource	→ batch[] →	Batch	1 link visible to multiple batches
Progress	→ student + video →	User + Video	Tracks 1 student's watch % on 1 video
EmailLog	→ sentBy →	User	Who sent which email

Tech Stack — Poori Technology List

FRONTEND	
Package/Tool	Kaam kya karta hai
React + Vite	Fast SPA bundler
React Router v6	Multi-page navigation
Axios	HTTP client for API calls
Recharts	Dashboard charts + graphs
React Player	Video playback component

BACKEND	
Package/Tool	Kaam kya karta hai
Node.js + Express	Server + routing
MongoDB + Mongoose	Database + ODM
JWT + bcryptjs	Auth + password hashing
Multer	File upload middleware
Nodemailer	Email sending

INFRASTRUCTURE	
Package/Tool	Kaam kya karta hai
Cloudinary	Video/image cloud storage
MongoDB Atlas	Cloud database
Render.com	Backend free deployment
Netlify	Frontend free deployment
Gmail SMTP	Email delivery service

FOLDER STRUCTURE — Poora Project Layout

```
learnkins-portal/
  ■■■ frontend/ ← React + Vite
  ■ ■■■ src/
  ■ ■■■ components/ ← Reusable: Button, Card, Modal, Badge
  ■ ■■■ pages/
  ■ ■■■ admin/
  ■ ■■■ ■■■ Dashboard.jsx
  ■ ■■■ ■■■ UploadVideo.jsx
  ■ ■■■ ■■■ ManageVideos.jsx
  ■ ■■■ ■■■ Resources.jsx
  ■ ■■■ ■■■ CreateStudent.jsx
  ■ ■■■ ■■■ StudentList.jsx
  ■ ■■■ ■■■ SendEmail.jsx
  ■ ■■■ ■■■ BatchManager.jsx
  ■ ■■■ ■■■ student/
  ■ ■■■ ■■■ Dashboard.jsx
  ■ ■■■ ■■■ VideoLibrary.jsx
  ■ ■■■ ■■■ WatchVideo.jsx
  ■ ■■■ ■■■ Resources.jsx
  ■ ■■■ ■■■ MyProgress.jsx
  ■ ■■■ context/AuthContext.jsx
  ■ ■■■ hooks/useFetch.js
  ■ ■■■ utils/api.js ← Axios instance + interceptors
  ■ ■■■ App.jsx ← Routes + PrivateRoute
  ■ ■■■ .env
  ■
  ■■■ backend/ ← Express + Node
  ■■■ routes/
  ■ ■■■ auth.js
  ■ ■■■ videos.js
  ■ ■■■ resources.js
  ■ ■■■ students.js
  ■ ■■■ batches.js
  ■ ■■■ email.js
  ■ ■■■ analytics.js
  ■■■ controllers/ ← Business logic (1 per route)
  ■■■ models/ ← Mongoose schemas
  ■■■ middleware/
  ■ ■■■ auth.js ← JWT verify + role check
  ■ ■■■ upload.js ← Multer + Cloudinary stream
  ■ ■■■ rateLimiter.js
  ■■■ utils/
  ■ ■■■ sendEmail.js
  ■ ■■■ cloudinary.js
  ■■■ server.js
  ■■■ .env
```

Environment Variables — .env Setup Guide

Ye variables kisi ko bhi mat dikhao! GitHub pe push mat karo! .gitignore mein .env daalo.

BACKEND .ENV

```
PORT=5000
MONGODB_URI=mongodb+srv://user:pass@cluster.mongodb.net/learnkins
JWT_SECRET=koi_bhi_random_64_char_string
JWT_EXPIRE=7d
CLOUDINARY_NAME=your_cloud_name
CLOUDINARY_API_KEY=your_api_key
CLOUDINARY_API_SECRET=your_api_secret
EMAIL_HOST=smtp.gmail.com
EMAIL_PORT=587
EMAIL_USER=learnkins@gmail.com
EMAIL_PASS=your_gmail_app_password
NODE_ENV=development
```

FRONTEND .ENV

```
VITE_API_URL=http://localhost:5000
# Production mein:
VITE_API_URL=https://learnkins-backend.onrender.com
```

Deployment Checklist — Live Karne Se Pehle

Backend — Render.com

- ☐ GitHub pe backend code push karo
- ☐ Render.com pe 'New Web Service' → GitHub repo connect karo
- ☐ Build Command: npm install
- ☐ Start Command: node server.js
- ☐ Environment tab mein sab .env variables daalo
- ☐ Deploy karo — URL milegi (<https://learnkins-backend.onrender.com>)

Frontend — Netlify

- ☐ VITE_API_URL ko Render ka URL se update karo frontend .env.production mein
- ☐ npm run build karo — dist/ folder banega
- ☐ Netlify.com → 'New Site' → dist/ folder drag karo
- ☐ Ya GitHub connect karo auto-deploy ke liye
- ☐ public/_redirects file banao: /* /index.html 200
- ☐ Environment variables mein VITE_API_URL set karo

Final Checks

- ☐ Live URL pe login test karo — admin aur student dono
- ☐ Video upload test karo — Cloudinary pe save ho raha hai?
- ☐ Email send test karo — Gmail SMTP kaam kar raha hai?
- ☐ Mobile pe open karo — responsive hai?
- ☐ README.md likho GitHub ke liye
- ☐ Admin credentials securely store karo

Quick Reference — Sab Kuch Ek Jagah

Tool	URL	Kab Use Karo	From
VS Code	code.visualstudio.com	Code editor — sab kuch yahan likhenge	All
Node.js LTS	nodejs.org	JavaScript server pe chalane ke liye	Day 1+
MongoDB Atlas	cloud.mongodb.com	Free cloud database	Day 1+
Cloudinary	cloudinary.com	Video + image cloud storage	Day 4+
Postman	postman.com	APIs test karne ke liye	Day 2+
Netlify	netlify.com	Frontend free deployment	Day 15+
Render	render.com	Backend free deployment	Day 15+
GitHub	github.com	Code version control + portfolio	All
React Docs	react.dev	Official React documentation	Day 1+
Mongoose Docs	mongoosejs.com/docs	Mongoose official guide	Day 1+
Express Docs	expressjs.com	Express.js documentation	Day 1+
JWT.io	jwt.io	JWT token debugger + docs	Day 1+